5 Sets of ICSE Class 10 Chemistry - PYQ Based Revision Material

Set 1

30 Most Important Numericals

1. **(2024)** An organic compound has a vapour density of 22. The molecular formula of the organic compound is: (Atomic weight: C = 12, H = 1) [Ans: C_3H_8]

2. **(2024)** 40 cm^3 of methane CH_4 is reacted with 60 cm^3 of oxygen. $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$. All volume is measured at room temperature. What is the total volume of the gases remaining at the end of the reaction? [Ans: 40 cm^3]

3. **(2015)** Complete the calculation: Calculate the mass of Calcium that will contain the same number of atoms as are present in 3.2 gm of Sulphur. (Atomic masses: S = 32, Ca = 40) [Ans: 4 g]

4. **(2015)** If 6 litres of hydrogen and 4 litres of chlorine are mixed and exploded and if water is added to the gases formed, find the volume of the residual gas. [Ans: 2 litres]

5. **(2015)** If the empirical formula of a compound is CH and it has a vapour density of 13, find the molecular formula of the compound. [Ans: C_2H_2]

6. **(2016)** A gas of mass 32 gms has a volume of 20 litres at S.T.P. Calculate the gram molecular weight of the gas. [Ans: 35.84 g]

7. **(2016)** How much Calcium oxide is formed when 82g of calcium nitrate is heated? $2Ca(NO_3)_2 \rightarrow 2CaO + 4NO_2 + O_2$ (Ca = 40, N = 14, O = 16) [Ans: 28 g of CaO]

8. **(2016)** Find the volume of nitrogen dioxide evolved from the above reaction. [Ans: 22.4 L at STP]

9. **(2019)** The number of moles in 12 g of oxygen gas [O = 16] [Ans: 0.375 moles]

10. **(2019)** The weight of 10^{22} atoms of carbon. [C = 12, Avogadro's No. = 6×10^{23}] [Ans: 0.2 g]

11. **(2019)** Molecular formula of a compound is $C_6H_{18}O_3$. Find its empirical formula. [Ans: C_2H_6O]

12. **(2020)** The volume occupied by 80 g of carbon dioxide at STP. (Atomic mass: C = 12, O = 16) [Ans: 40.72 L]

13. **(2020)** Calculate the number of molecules in 4.4 gm of CO_2. [Ans: 6.022×10^{22} molecules]

14. **(2022 Sem 1)** If Relative Molecular Mass of Butane (C_4H_{10}) is 58 then, its vapour density will be: [Ans: 29]

15. **(2022 Sem 1)** The percentage of hydrogen present in NaOH is: (Relative Molecular Mass of NaOH = 40, At. Wt. of H = 1) [Ans: 2.5%]

16. **(2022 Sem 1)** If the empirical mass of the formula PQ_2 is 10 and the Relative Molecular Mass is 30, then the molecular formula will be: [Ans: P_3Q_6]

17. **(2022 Sem 1)** If the empirical formula of a compound is CH and its vapour density is 13, then its molecular formula will be: (At. Wt. C = 12, H = 1) [Ans: C_2H_2]

18. **(2023)** The vapour density of CH_3OH is (At. Wt. C = 12, H = 1, O = 16) [Ans: 16]

19. **(2023)** Ethane burns in oxygen according to the chemical equation: $2C_2H_6 + 7O_2 \rightarrow 4CO_2 + 6H_2O$. If 80 ml of ethane is burnt in 300 ml of oxygen, find the composition of the resultant gaseous mixture when measured at room temperature. [Ans: 160 ml CO_2 and 20 ml O_2]

20. **(2023)** The Empirical formula of an organic compound is $CHCl_2$. If its relative molecular mass is 168, what is its molecular formula? (At. Wt. C = 12, H = 1, Cl = 35.5) [Ans: $C_2H_2Cl_4$]

21. **(2023)** Calculate the percentage of phosphate in the fertilizer super phosphate $Ca(H_2PO_4)_2$ correct to 1 decimal point. (At. Wt. H = 1, P = 31, O = 16, Ca = 40) [Ans: 26.5%]

22. **(2023)** Write the empirical formula of C_8H_{18}. [Ans: C_4H_9]

23. **(2018)** The percentage composition of a gas is: Nitrogen 82.35%, Hydrogen 17.64%. Find the empirical formula of the gas. [N = 14, H = 1] [Ans: NH_3]

24. **(2018)** Aluminum carbide reacts with water: $Al_4C_3 + 12H_2O \rightarrow 4Al(OH)_3 + 3CH_4$. What mass of aluminum hydroxide is formed from 12 g of aluminum carbide? (Relative molecular weight of Al_4C_3 = 144; $Al(OH)_3$ = 78) [Ans: 26 g]

25. **(2018)** What volume of methane at s.t.p. is obtained from 12 g of aluminum carbide? [Ans: 5.6 L CH_4]

26. **(2018)** If 150 cc of gas A contains X molecules, how many molecules of gas B will be present in 75 cc of B? The gases A and B are under the same conditions of temperature and pressure. [Ans: X/2 molecules]

27. **(2019)** Copper sulphate solution reacts with sodium hydroxide solution: $2NaOH + CuSO_4 \rightarrow Na_2SO_4 + Cu(OH)_2$. What mass of copper hydroxide is precipitated by using 200 gm of sodium hydroxide? [H = 1, O = 16, Na = 23, S = 32, Cu = 64] [Ans: 245 g]

28. **(2019)** Find the empirical formula and molecular formula of an organic compound from the data given below: C = 75.92%, H = 6.32%, N = 17.76%. The vapour density of the compound is 39.5. [C = 12, H = 1, N = 14] [Ans: C_5H_5N]

29. **(2020)** The amount of each reactant required to produce 750 ml of carbon dioxide, when two volumes of carbon monoxide combine with one volume of oxygen to produce two volumes of carbon dioxide. $2CO + O_2 \rightarrow 2CO_2$ [Ans: 750 ml CO, 375 ml O_2]

30. **(2016)** A gas cylinder contains 12×10^{24} molecules of oxygen gas. If Avogadro's number is 6×10^{23}, calculate: (i) the mass of oxygen present in the cylinder. (ii) the volume of oxygen at S.T.P. present in the cylinder. [O = 16] [Ans: 640 g, 448 L]

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20 Formulas

S.No.	Formula	Description
1	Molecular mass = 2 × Vapour density	Relation between molecular mass and vapour density
2	Number of moles = Given mass / Molar mass	Mole concept
3	Number of moles = Volume at STP (L) / 22.4 L	Mole-volume relationship at STP
4	Number of particles = Number of moles × Avogadro's number	Relation between moles and particles
5	Empirical formula = Simplest whole number ratio of atoms	Empirical formula calculation
6	Molecular formula = n × Empirical formula	Where n = Molecular mass / Empirical formula mass
7	% of element = (Mass of element / Molecular mass) × 100	Percentage composition
8	% of water of crystallization = (Mass of water / Total mass) × 100	Water of crystallization
9	$V_1/T_1 = V_2/T_2$	Charles' Law
10	$P_1V_1 = P_2V_2$	Boyle's Law
11	$P_1V_1/T_1 = P_2V_2/T_2$	Combined gas equation
12	$V_1/n_1 = V_2/n_2$	Avogadro's Law
13	$Na + 2H_2O \rightarrow 2NaOH + H_2$	Reaction of sodium with water
14	$2Al + 2NaOH + 2H_2O \rightarrow 2NaAlO_2 + 3H_2$	Reaction of aluminium with alkali
15	$Zn + 2HCl \rightarrow ZnCl_2 + H_2$	Reaction of zinc with dilute acid
16	$2KClO_3 \rightarrow 2KCl + 3O_2$	Thermal decomposition of $KClO_3$
17	$2H_2O \rightarrow 2H_2 + O_2$	Electrolysis of water
18	$CuO + H_2 \rightarrow Cu + H_2O$	Reduction of copper oxide
19	$2Mg + O_2 \rightarrow 2MgO$	Combination reaction

| 20 | $NH_3 + HCl \rightarrow NH_4Cl$ | Reaction of ammonia with HCl |

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10 Diagrams

1. **(2020)** Electron dot diagram of water molecule (H_2O)
2. **(2019)** Electron dot structure of nitrogen molecule (N_2)
3. **(2019)** Electron dot structure of sodium chloride (NaCl)
4. **(2019)** Electron dot structure of ammonium ion (NH_4^+)
5. **(2020)** Structural formula of 2,2-dimethylpentane
6. **(2020)** Structural formula of methanol
7. **(2020)** Structural formula of isopropane
8. **(2022 Sem 2)** Structural diagram of pentanal
9. **(2022 Sem 2)** Structural diagram of propanol
10. **(2022 Sem 2)** Structural diagram of 2-butene

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10 Definitions

1. **Ionisation Potential (2016)**: The energy required to remove an electron from a neutral isolated gaseous atom and convert it into a positively charged gaseous ion.

2. **Electron Affinity (2016)**: The amount of energy released when an atom in the gaseous state accepts an electron to form an anion.

3. **Electronegativity (2016)**: The tendency of an atom to attract electrons to itself when combined in a compound.

4. **Catenation (2018)**: The property by which carbon links with itself to form a long chain.

5. **Isomerism (2019)**: The phenomenon where compounds have the same molecular formula but different structural formulae.

6. **Empirical Formula (2019)**: The formula that represents the simplest ratio of the various elements present in one molecule of the compound.

7. **Ore (2022 Sem 2)**: A naturally occurring mineral having a high concentration of a certain element.

8. **Calcination (2019)**: The process by which certain ores, specially carbonates, are converted to oxides in the absence of air.

9. **Covalent Bond (2019)**: The bond in which the electrons are shared equally between the combining atoms.

10. **Homologous Series (2023)**: A group of organic compounds where the successive members follow a regular structural pattern and differ by a CH_2 group.

Set 2

30 Most Important Numericals

1. **(2024)** A compound X and Y has the empirical formula XY_2. Its vapour density is equal to its empirical formula weight. Determine its molecular formula. [Ans: X_2Y_4]

2. **(2015)** 4.5 moles of calcium carbonate are reacted with dilute hydrochloric acid. $CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$. What is the mass of 4.5 moles of calcium carbonate? (Relative molecular mass of calcium carbonate is 100) [Ans: 450 g]

3. **(2015)** What is the volume of carbon dioxide liberated at STP from 4.5 moles of calcium carbonate? [Ans: 100.8 L]

4. **(2015)** What mass of calcium chloride is formed from 4.5 moles of calcium carbonate? (Relative molecular mass of calcium chloride is 111) [Ans: 499.5 g]

5. **(2015)** How many moles of HCl are used in this reaction with 4.5 moles of calcium carbonate? [Ans: 9 moles]

6. **(2016)** A gas cylinder contains 12×10^{24} molecules of oxygen gas. If Avogadro's number is 6×10^{23}, calculate the mass of oxygen present. [O = 16] [Ans: 640 g]

7. **(2016)** Calculate the volume of oxygen at S.T.P. present in the cylinder containing 12×10^{24} molecules. [Ans: 448 L]

8. **(2016)** A gaseous hydrocarbon contains 82.76% of carbon. Given that its vapour density is 29, find its molecular formula. [C = 12, H = 1] [Ans: C_4H_{10}]

9. **(2016)** The equation $4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$ represents the catalytic oxidation of ammonia. If 100 cm^3 of ammonia is used, calculate the volume of oxygen required to oxidise the ammonia completely. [Ans: 125 cm^3]

10. **(2019)** Calculate: The number of moles in 12 g of oxygen gas [O = 16] [Ans: 0.375 moles]

11. **(2019)** Calculate the weight of 10^{22} atoms of carbon. [C = 12, Avogadro's No. = 6×10^{23}] [Ans: 0.2 g]

12. **(2020)** Calculate the number of moles in 11 gm of nitrogen gas. (Atomic mass of N = 14) [Ans: 0.39 moles]

13. **(2022 Sem 1)** The percentage of hydrogen present in NaOH is: (Relative Molecular Mass of NaOH = 40, At. Wt. of H = 1) [Ans: 2.5%]

14. **(2022 Sem 1)** If the empirical mass of the formula PQ_2 is 10 and the Relative Molecular Mass is 30, then the molecular formula will be: [Ans: P_3Q_6]

15. **(2023)** Ethane burns in oxygen: $2C_2H_6 + 7O_2 \rightarrow 4CO_2 + 6H_2O$. If 80 ml of ethane is burnt in 300 ml of oxygen, find the volume of CO_2 produced. [Ans: 160 ml]

16. **(2023)** Find the volume of unused oxygen from the above reaction. [Ans: 20 ml]

17. **(2018)** Aluminum carbide reacts with water: $Al_4C_3 + 12H_2O \rightarrow 4Al(OH)_3 + 3CH_4$. What mass of aluminum hydroxide is formed from 12 g of aluminum carbide? [Ans: 26 g]

18. **(2018)** What volume of methane at s.t.p. is obtained from 12 g of aluminum carbide? [Ans: 5.6 L]

19. **(2018)** If 150 cc of gas A contains X molecules, how many molecules of gas B will be present in 75 cc of B? [Ans: X/2 molecules]

20. **(2019)** A compound has the molecular formula $C_6H_{18}O_3$. Find its empirical formula. [Ans: C_2H_6O]

21. **(2019)** Find the empirical formula of a compound with C = 75.92%, H = 6.32%, N = 17.76%. [Ans: C_5H_5N]

22. **(2019)** If the vapour density of the above compound is 39.5, find its molecular formula. [Ans: C_5H_5N]

23. **(2020)** State the volume occupied by 40 gm of methane at STP, if its vapour density (V.D.) is 8. [Ans: 56 L]

24. **(2020)** Calculate the number of moles present in 160 gm of NaOH. (Atomic mass: Na = 23, H = 1, O = 16) [Ans: 4 moles]

25. **(2023)** Calculate the percentage of phosphate in $Ca(H_2PO_4)_2$. [Ans: 26.5%]

26. **(2023)** Write the empirical formula of C_8H_{18}. [Ans: C_4H_9]

27. **(2016)** Calculate the mass of 0.1 mole of carbon dioxide. [C = 12, O = 16] [Ans: 4.4 g]

28. **(2016)** Calculate the mass of 10^{22} atoms of sulphur. [S = 32, Avogadro's Number = 6×10^{23}] [Ans: 0.533 g]

29. **(2015)** 67.2 litres of hydrogen combines with 44.8 litres of nitrogen to form ammonia: $N_2 + 3H_2 \rightarrow 2NH_3$. Calculate the volume of ammonia produced. [Ans: 44.8 L]

30. **(2015)** The mass of 5.6 dm^3 of a certain gas at STP is 12.0 g. Calculate the relative molecular mass of the gas. [Ans: 48]

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20 Formulas

S.No.	Formula	Description
1	$NaCl + H_2SO_4$ (conc.) $\rightarrow NaHSO_4 + HCl$	Lab preparation of HCl (below 200°C)
2	$2NaCl + H_2SO_4$ (conc.) $\rightarrow Na_2SO_4 + 2HCl$	Lab preparation of HCl (above 200°C)
3	$2NH_4Cl + Ca(OH)_2 \rightarrow CaCl_2 + 2H_2O + 2NH_3$	Lab preparation of ammonia
4	$4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$	Catalytic oxidation of ammonia
5	$NH_3 + HCl \rightarrow NH_4Cl$	Reaction of ammonia with HCl
6	$2NH_3 + 3Cl_2 \rightarrow N_2 + 6HCl$	Reaction of ammonia with chlorine
7	$NH_3 + 3Cl_2$ (excess) $\rightarrow NCl_3 + 3HCl$	Reaction of ammonia with excess chlorine
8	$3CuO + 2NH_3 \rightarrow 3Cu + 3H_2O + N_2$	Ammonia as reducing agent
9	$CaC_2 + 2H_2O \rightarrow Ca(OH)_2 + C_2H_2$	Preparation of ethyne
10	C_2H_5OH (alc) + conc. H_2SO_4 at 170°C $\rightarrow C_2H_4 + H_2O$	Preparation of ethene
11	$C_2H_4 + H_2O$ (steam) $\rightarrow C_2H_5OH$	Hydration of ethene
12	$CH_3COONa + NaOH$ (soda lime) $\rightarrow CH_4 + Na_2CO_3$	Preparation of methane
13	$C_2H_5Br + KOH$ (alc) $\rightarrow C_2H_4 + KBr + H_2O$	Dehydrohalogenation
14	$C_2H_5Br + KOH$ (aq) $\rightarrow C_2H_5OH + KBr$	Hydrolysis
15	$C_2H_4 + H_2$ (Ni catalyst) $\rightarrow C_2H_6$	Hydrogenation
16	$C_2H_4 + Br_2 \rightarrow C_2H_4Br_2$	Addition reaction
17	$CH_4 + Cl_2$ (sunlight) $\rightarrow CH_3Cl + HCl$	Substitution reaction
18	$2Al_2O_3$ (electrolysis) $\rightarrow 4Al + 3O_2$	Extraction of aluminium
19	$2Al(OH)_3 \rightarrow Al_2O_3 + 3H_2O$	Decomposition of aluminium hydroxide
20	$Al_2O_3 + 2NaOH \rightarrow 2NaAlO_2 + H_2O$	Bayer's process

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10 Diagrams

1. **(2023)** Electron dot structure of ammonium ion (NH_4^+)

2. **(2023)** Electron dot structure of nitrogen molecule (N_2)
3. **(2019)** Electron dot diagram of methane (CH_4)
4. **(2022 Sem 2)** Structural formula of 2-butene
5. **(2020)** Structural formula of 2,2-dimethylpentane
6. **(2023)** Sketch of Brown Ring Test
7. **(2022 Sem 2)** Diagram of electroplating setup
8. **(2018)** Diagram for laboratory preparation of HCl gas
9. **(2019)** Extract of Periodic Table with elements labelled
10. **(2020)** Diagram showing electrolysis of alumina (Hall-Heroult process)

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10 Definitions

1. **Acid (2019)**: The substance that releases hydronium ion as the only positive ion when dissolved in water.

2. **Base (IX Syllabus)**: A substance that releases hydroxyl ions (OH^-) as the only negative ions when dissolved in water.

3. **Salt (IX Syllabus)**: A compound formed by partial or complete replacement of the hydrogen ion of an acid by a metal.

4. **Electrolyte (2019)**: A substance that conducts electricity in aqueous solution or molten state due to the presence of ions.

5. **Strong Electrolyte (2019)**: An electrolyte that undergoes almost complete ionisation and allows large current to pass through.

6. **Weak Electrolyte (2019)**: An electrolyte that undergoes partial ionisation and allows only a weak current to pass through.

7. **Non-electrolyte (2019)**: A substance that consists only of molecules and does not conduct electricity.

8. **Roasting (2019)**: The process of heating an ore in a regular supply of air to convert sulphide ores to oxides.

9. **Calcination (2019)**: The process of heating an ore in limited supply or absence of air to convert carbonate ores to oxides.

10. **Alloy (2020)**: The substance prepared by adding other metals to a base metal in appropriate proportions to obtain certain desirable properties.

Set 3

30 Most Important Numericals

1. **(2024)** 40 cm^3 of methane is reacted with 60 cm^3 of oxygen. $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$. What is the total volume of gases remaining? [Ans: 40 cm^3]

2. **(2015)** If 6 litres of hydrogen and 4 litres of chlorine are mixed and exploded, find the volume of residual gas after adding water. [Ans: 2 litres]

3. **(2016)** A gas of mass 32 g has a volume of 20 litres at STP. Calculate the gram molecular weight. [Ans: 35.84 g]

4. **(2016)** 82 g of calcium nitrate is heated: $2Ca(NO_3)_2 \rightarrow 2CaO + 4NO_2 + O_2$. Find mass of CaO formed. [Ans: 28 g]

5. **(2016)** Find the volume of NO_2 evolved at STP from the above reaction. [Ans: 22.4 L]

6. **(2019)** Calculate the number of moles in 12 g of oxygen gas. [Ans: 0.375 moles]

7. **(2019)** Calculate the weight of 10^{22} atoms of carbon. [Ans: 0.2 g]

8. **(2019)** Molecular formula of a compound is $C_6H_{18}O_3$. Find its empirical formula. [Ans: C_2H_6O]

9. **(2020)** Calculate the number of molecules in 4.4 gm of CO_2. [Ans: 6.022×10^{22} molecules]

10. **(2020)** The volume occupied by 80 g of carbon dioxide at STP. [Ans: 40.72 L]

11. **(2022 Sem 1)** If Relative Molecular Mass of Butane is 58, its vapour density is: [Ans: 29]

12. **(2022 Sem 1)** The percentage of hydrogen in NaOH is: [Ans: 2.5%]

13. **(2022 Sem 1)** If the empirical mass of PQ_2 is 10 and RMM is 30, molecular formula is: [Ans: P_3Q_6]

14. **(2023)** The vapour density of CH_3OH is: [Ans: 16]

15. **(2023)** If 80 ml ethane is burnt in 300 ml oxygen, find composition of mixture. [Ans: 160 ml CO_2, 20 ml O_2]

16. **(2023)** Empirical formula $CHCl_2$, RMM = 168. Find molecular formula. [Ans: $C_2H_2Cl_4$]

17. **(2023)** Calculate percentage of phosphate in $Ca(H_2PO_4)_2$. [Ans: 26.5%]

18. **(2023)** Write empirical formula of C_8H_{18}. [Ans: C_4H_9]

19. **(2018)** Find empirical formula of gas with N = 82.35%, H = 17.64%. [Ans: NH_3]

20. **(2018)** 12 g Al_4C_3 gives how much $Al(OH)_3$? (Al_4C_3 = 144, $Al(OH)_3$ = 78) [Ans: 26 g]

21. **(2018)** Volume of CH_4 at STP from 12 g Al_4C_3. [Ans: 5.6 L]

22. **(2018)** 150 cc gas A has X molecules. How many molecules in 75 cc gas B? [Ans: X/2]

23. **(2019)** 200 g NaOH gives how much $Cu(OH)_2$? $2NaOH + CuSO_4 \rightarrow Na_2SO_4 + Cu(OH)_2$ [Ans: 245 g]

24. **(2019)** Empirical formula of compound with C = 75.92%, H = 6.32%, N = 17.76%. [Ans: C_5H_5N]

25. **(2019)** If VD = 39.5, find molecular formula of above compound. [Ans: C_5H_5N]

26. **(2020)** Volume of 40 g methane at STP (VD = 8). [Ans: 56 L]

27. **(2020)** Number of moles in 160 g NaOH. [Ans: 4 moles]

28. **(2016)** 100 cm^3 ammonia oxidized: $4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$. Find volume of oxygen required. [Ans: 125 cm^3]

29. **(2016)** Find molecular formula of hydrocarbon with 82.76% C and VD = 29. [Ans: C_4H_{10}]

30. **(2015)** 67.2 L H_2 combines with 44.8 L N_2 to form NH_3. Find volume of NH_3 produced. [Ans: 44.8 L]

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20 Formulas

S.No.	Formula	Description
1	$2KClO_3$ (MnO_2) $\rightarrow 2KCl + 3O_2$	Lab preparation of oxygen
2	$2H_2O$ (electrolysis) $\rightarrow 2H_2 + O_2$	Electrolysis of water
3	$Zn + H_2SO_4$ (dil) $\rightarrow ZnSO_4 + H_2$	Preparation of hydrogen
4	$2Al + 6HCl \rightarrow 2AlCl_3 + 3H_2$	Reaction of aluminium with acid
5	$Fe + 2HCl \rightarrow FeCl_2 + H_2$	Reaction of iron with acid

6	$2Na + 2H_2O \rightarrow 2NaOH + H_2$	Reaction of sodium with water
7	$Ca + 2H_2O \rightarrow Ca(OH)_2 + H_2$	Reaction of calcium with water
8	$2K + 2H_2O \rightarrow 2KOH + H_2$	Reaction of potassium with water
9	$Mg + H_2SO_4 \rightarrow MgSO_4 + H_2$	Reaction of magnesium with acid
10	$2Al + 2NaOH + 2H_2O \rightarrow 2NaAlO_2 + 3H_2$	Reaction of aluminium with alkali
11	$Zn + 2NaOH \rightarrow Na_2ZnO_2 + H_2$	Reaction of zinc with alkali
12	$Pb + 2NaOH \rightarrow Na_2PbO_2 + H_2$	Reaction of lead with alkali
13	$C + 2H_2SO_4$ (conc) $\rightarrow CO_2 + 2SO_2 + 2H_2O$	Oxidation of carbon
14	$S + 2H_2SO_4$ (conc) $\rightarrow 3SO_2 + 2H_2O$	Oxidation of sulphur
15	$Cu + 4HNO_3$ (conc) $\rightarrow Cu(NO_3)_2 + 2NO_2 + 2H_2O$	Reaction of copper with conc. HNO_3
16	$3Cu + 8HNO_3$ (dil) $\rightarrow 3Cu(NO_3)_2 + 2NO + 4H_2O$	Reaction of copper with dil. HNO_3
17	$2Pb(NO_3)_2$ (heat) $\rightarrow 2PbO + 4NO_2 + O_2$	Thermal decomposition of lead nitrate
18	$2Cu(NO_3)_2$ (heat) $\rightarrow 2CuO + 4NO_2 + O_2$	Thermal decomposition of copper nitrate
19	$2Zn(NO_3)_2$ (heat) $\rightarrow 2ZnO + 4NO_2 + O_2$	Thermal decomposition of zinc nitrate
20	NH_4Cl (heat) $\rightleftharpoons NH_3 + HCl$	Thermal dissociation of ammonium chloride

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10 Diagrams

1. **(2023)** Electron dot structure of ammonium ion
2. **(2019)** Electron dot structure of nitrogen molecule
3. **(2019)** Electron dot structure of sodium chloride
4. **(2020)** Structural formula of methanol
5. **(2022 Sem 2)** Structural formula of pentanal
6. **(2022 Sem 2)** Structural formula of propanol
7. **(2020)** Structural formula of 2,2-dimethylpentane
8. **(2022 Sem 2)** Structural formula of 2-butene
9. **(2019)** Electron dot structure of methane
10. **(2023)** Brown Ring Test diagram

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10 Definitions

1. **Ionisation (2016)**: Formation of ions from molecules.

2. **Electrolysis (2016)**: The process of decomposition of an electrolyte by passing electric current through its molten or aqueous state.

3. **Electrolyte (2016)**: A substance that conducts electricity in molten or aqueous state and undergoes chemical decomposition.

4. **Electrode (2016)**: The two metal plates or wires through which the current enters and leaves the electrolytic cell.

5. **Anode (2016)**: The electrode connected to the positive terminal of the battery where oxidation takes place.

6. **Cathode (2016)**: The electrode connected to the negative terminal of the battery where reduction takes place.

7. **Cation (2016)**: A positively charged ion that migrates towards the cathode during electrolysis.

8. **Anion (2016)**: A negatively charged ion that migrates towards the anode during electrolysis.

9. **Oxidation (2016)**: The loss of electrons from an atom or ion.

10. **Reduction (2016)**: The gain of electrons by an atom or ion.

Set 4

30 Most Important Numericals

1. **(2024)** A compound has vapour density 22. Find its molecular formula if it contains C and H only. [Ans: C_3H_8]

2. **(2024)** 40 cm^3 CH_4 reacts with 60 cm^3 O_2. Find total volume of gases remaining. [Ans: 40 cm^3]

3. **(2015)** Find mass of Ca containing same atoms as 3.2 g S. [Ans: 4 g]

4. **(2015)** 6 L H_2 + 4 L Cl_2 exploded. Find residual gas after adding water. [Ans: 2 L]

5. **(2015)** Empirical formula CH, VD = 13. Find molecular formula. [Ans: C_2H_2]

6. **(2016)** 32 g gas occupies 20 L at STP. Find gram molecular weight. [Ans: 35.84 g]

7. **(2016)** 82 g $Ca(NO_3)_2$ heated. Find mass of CaO formed. [Ans: 28 g]

8. **(2016)** Find volume of NO_2 evolved at STP from above. [Ans: 22.4 L]

9. **(2019)** Number of moles in 12 g O_2. [Ans: 0.375 moles]

10. **(2019)** Weight of 10^{22} atoms of C. [Ans: 0.2 g]

11. **(2019)** Empirical formula of $C_6H_{18}O_3$. [Ans: C_2H_6O]

12. **(2020)** Number of molecules in 4.4 g CO_2. [Ans: 6.022×10^{22}]

13. **(2020)** Volume of 80 g CO_2 at STP. [Ans: 40.72 L]

14. **(2022 Sem 1)** Vapour density of C_4H_{10} (RMM = 58). [Ans: 29]

15. **(2022 Sem 1)** % H in NaOH. [Ans: 2.5%]

16. **(2022 Sem 1)** If empirical mass of PQ_2 = 10, RMM = 30, find molecular formula. [Ans: P_3Q_6]

17. **(2022 Sem 1)** Empirical formula CH, VD = 13. Find molecular formula. [Ans: C_2H_2]

18. **(2023)** Vapour density of CH_3OH. [Ans: 16]

19. **(2023)** 80 ml C_2H_6 + 300 ml O_2. Find volume of CO_2 produced. [Ans: 160 ml]

20. **(2023)** Find volume of unused O_2 from above. [Ans: 20 ml]

21. **(2023)** Empirical formula $CHCl_2$, RMM = 168. Find molecular formula. [Ans: $C_2H_2Cl_4$]

22. **(2023)** % phosphate in $Ca(H_2PO_4)_2$. [Ans: 26.5%]

23. **(2023)** Empirical formula of C_8H_{18}. [Ans: C_4H_9]

24. **(2018)** 12 g Al_4C_3 gives how much $Al(OH)_3$? [Ans: 26 g]

25. **(2018)** Volume of CH_4 at STP from 12 g Al_4C_3. [Ans: 5.6 L]

26. **(2018)** 150 cc gas A has X molecules. Molecules in 75 cc gas B? [Ans: X/2]

27. **(2019)** 200 g NaOH gives how much $Cu(OH)_2$? [Ans: 245 g]

28. **(2019)** Empirical formula of compound with C = 75.92%, H = 6.32%, N = 17.76%. [Ans: C_5H_5N]

29. **(2019)** Molecular formula if VD = 39.5. [Ans: C_5H_5N]

30. **(2020)** Volume of 40 g CH_4 at STP (VD = 8). [Ans: 56 L]

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20 Formulas

S.No.	Formula	Description
1	$CuO + H_2 \rightarrow Cu + H_2O$	Reduction of copper oxide
2	$PbO + H_2 \rightarrow Pb + H_2O$	Reduction of lead oxide
3	$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$	Reduction of iron oxide
4	$ZnO + C \rightarrow Zn + CO$	Reduction of zinc oxide
5	$2Al_2O_3$ (electrolysis) $\rightarrow 4Al + 3O_2$	Extraction of aluminium
6	$2NaCl$ (electrolysis) $\rightarrow 2Na + Cl_2$	Extraction of sodium
7	$2NaCl + 2H_2O$ (electrolysis) $\rightarrow 2NaOH + H_2 + Cl_2$	Chlor-alkali process
8	$2Ag + 2H_2SO_4$ (conc) $\rightarrow Ag_2SO_4 + SO_2 + 2H_2O$	Reaction of silver with conc. H_2SO_4
9	$2Ag + 2HNO_3$ (conc) $\rightarrow 2AgNO_3 + H_2O + NO_2$	Reaction of silver with conc. HNO_3
10	$2Al + 3Cl_2 \rightarrow 2AlCl_3$	Direct combination
11	$2Fe + 3Cl_2 \rightarrow 2FeCl_3$	Direct combination
12	$Fe + S \rightarrow FeS$	Direct combination
13	$2Na + Cl_2 \rightarrow 2NaCl$	Direct combination
14	$CaO + H_2O \rightarrow Ca(OH)_2$	Slaking of lime
15	$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$	Test for CO_2
16	$CaCO_3$ (heat) $\rightarrow CaO + CO_2$	Thermal decomposition
17	$Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$	Reaction of carbonate with acid
18	$NaHCO_3 + HCl \rightarrow NaCl + H_2O + CO_2$	Reaction of bicarbonate with acid
19	$Na_2SO_3 + 2HCl \rightarrow 2NaCl + H_2O + SO_2$	Reaction of sulphite with acid
20	$Na_2S + 2HCl \rightarrow 2NaCl + H_2S$	Reaction of sulphide with acid

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10 Diagrams

1. **(2023)** Electron dot structure of nitrogen molecule
2. **(2019)** Electron dot structure of ammonium ion
3. **(2019)** Electron dot structure of sodium chloride
4. **(2020)** Structural formula of 2,2-dimethylpentane
5. **(2020)** Structural formula of methanol
6. **(2020)** Structural formula of isopropane
7. **(2022 Sem 2)** Structural diagram of pentanal
8. **(2022 Sem 2)** Structural diagram of propanol
9. **(2022 Sem 2)** Structural diagram of 2-butene
10. **(2023)** Diagram of Hall-Heroult process for aluminium extraction

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10 Definitions

1. **Mole (2016)**: The amount of substance which contains the same number of units as the number of atoms in carbon-12.

2. **Atomic Number (IX Syllabus)**: The number of protons present in the nucleus of an atom.

3. **Mass Number (IX Syllabus)**: The total number of protons and neutrons present in the nucleus of an atom.

4. **Isotopes (IX Syllabus)**: Atoms of the same element having the same atomic number but different mass numbers.

5. **Isobars (2018)**: Atoms of different elements having the same mass number but different atomic numbers.

6. **Valence Electrons (IX Syllabus)**: The electrons present in the outermost shell of an atom.

7. **Octet Rule (IX Syllabus)**: The tendency of atoms to achieve eight electrons in their valence shell to attain stable configuration.

8. **Electrovalent Bond (IX Syllabus)**: The bond formed by the complete transfer of electrons from one atom to another.

9. **Covalent Bond (IX Syllabus)**: The bond formed by the sharing of electrons between two atoms.

10. **Coordinate Bond (IX Syllabus)**: A special type of covalent bond in which both electrons of the shared pair come from the same atom.

Set 5

30 Most Important Numericals

1. **(2024)** An organic compound has vapour density 22. Find its molecular formula if it contains C and H only. [Ans: C_3H_8]

2. **(2015)** 4.5 moles $CaCO_3$ reacts with HCl. Find mass of $CaCO_3$. [Ans: 450 g]

3. **(2015)** Find volume of CO_2 at STP from above reaction. [Ans: 100.8 L]

4. **(2015)** Find mass of $CaCl_2$ formed (RMM = 111). [Ans: 499.5 g]

5. **(2015)** Find moles of HCl used. [Ans: 9 moles]

6. **(2016)** 12×10^{24} molecules O_2. Find mass of O_2. [Ans: 640 g]

7. **(2016)** Find volume of O_2 at STP. [Ans: 448 L]

8. **(2016)** Hydrocarbon with 82.76% C and VD = 29. Find molecular formula. [Ans: C_4H_{10}]

9. **(2016)** 100 cm^3 NH_3 oxidized. Find volume of O_2 required. [Ans: 125 cm^3]

10. **(2018)** Find empirical formula of gas with N = 82.35%, H = 17.64%. [Ans: NH_3]

11. **(2018)** 12 g Al_4C_3 gives how much $Al(OH)_3$? [Ans: 26 g]

12. **(2018)** Volume of CH_4 at STP from 12 g Al_4C_3. [Ans: 5.6 L]

13. **(2018)** 150 cc gas A has X molecules. Molecules in 75 cc gas B? [Ans: X/2]

14. **(2019)** 200 g NaOH gives how much $Cu(OH)_2$? [Ans: 245 g]

15. **(2019)** Empirical formula of compound with C = 75.92%, H = 6.32%, N = 17.76%. [Ans: C_5H_5N]

16. **(2019)** Molecular formula if VD = 39.5. [Ans: C_5H_5N]

17. **(2020)** Volume of 40 g CH_4 at STP (VD = 8). [Ans: 56 L]

18. **(2020)** Number of moles in 160 g NaOH. [Ans: 4 moles]

19. **(2020)** Calculate number of molecules in 4.4 g CO_2. [Ans: 6.022×10^{22}]

20. **(2020)** 750 ml CO_2 produced from CO + O_2. Find volume of CO and O_2 required. [Ans: 750 ml CO, 375 ml O_2]

21. **(2022 Sem 1)** If RMM of Butane = 58, find VD. [Ans: 29]

22. **(2022 Sem 1)** % H in NaOH. [Ans: 2.5%]

23. **(2022 Sem 1)** Empirical mass of PQ_2 = 10, RMM = 30. Find molecular formula. [Ans: P_3Q_6]

24. **(2022 Sem 1)** Empirical formula CH, VD = 13. Find molecular formula. [Ans: C_2H_2]

25. **(2023)** VD of CH_3OH. [Ans: 16]

26. **(2023)** 80 ml C_2H_6 + 300 ml O_2. Find composition. [Ans: 160 ml CO_2, 20 ml O_2]

27. **(2023)** Empirical formula $CHCl_2$, RMM = 168. Find molecular formula. [Ans: $C_2H_2Cl_4$]

28. **(2023)** % phosphate in $Ca(H_2PO_4)_2$. [Ans: 26.5%]

29. **(2023)** Empirical formula of C_8H_{18}. [Ans: C_4H_9]

30. **(2015)** 67.2 L H_2 + 44.8 L N_2. Find volume of NH_3 produced. [Ans: 44.8 L]

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20 Formulas

S.No.	Formula	Description
1	$2SO_2 + O_2 \rightleftharpoons 2SO_3$ (V_2O_5 catalyst, 450°C)	Contact process - step 1
2	$SO_3 + H_2SO_4 \rightarrow H_2S_2O_7$	Contact process - formation of oleum
3	$H_2S_2O_7 + H_2O \rightarrow 2H_2SO_4$	Contact process - dilution of oleum
4	$N_2 + 3H_2 \rightleftharpoons 2NH_3$ (Fe catalyst, 450-500°C, 200 atm)	Haber's process
5	$4NH_3 + 5O_2$ (Pt catalyst, 800°C) $\rightarrow 4NO + 6H_2O$	Ostwald's process - step 1
6	$2NO + O_2 \rightarrow 2NO_2$	Ostwald's process - step 2
7	$4NO_2 + O_2 + 2H_2O \rightarrow 4HNO_3$	Ostwald's process - step 3
8	$3NO_2 + H_2O \rightarrow 2HNO_3 + NO$	Alternative for step 3
9	$Ca_3(PO_4)_2 + 3H_2SO_4 \rightarrow 3CaSO_4 + 2H_3PO_4$	Manufacture of phosphoric acid
10	$Ca_3(PO_4)_2 + 2H_2SO_4 + 4H_2O \rightarrow Ca(H_2PO_4)_2 + 2CaSO_4 \cdot 2H_2O$	Manufacture of superphosphate
11	$CH_3COOH + C_2H_5OH \rightleftharpoons CH_3COOC_2H_5 + H_2O$	Esterification
12	$C_2H_5OH + Na \rightarrow C_2H_5ONa + ½H_2$	Reaction of alcohol with sodium
13	$C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$	Combustion of ethanol
14	$CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$	Neutralization of acetic acid
15	$2CH_3COOH + Mg \rightarrow (CH_3COO)_2Mg + H_2$	Reaction of acetic acid with magnesium
16	$CH_3COOH + Na_2CO_3 \rightarrow 2CH_3COONa + CO_2 + H_2O$	Reaction of acetic acid with carbonate
17	$C_{12}H_{22}O_{11}$ (conc. H_2SO_4) $\rightarrow 12C + 11H_2O$	Dehydration of sugar
18	$CuSO_4 \cdot 5H_2O$ (heat) $\rightarrow CuSO_4 + 5H_2O$	Dehydration of hydrated copper sulphate
19	$2Pb_3O_4 + 8HCl \rightarrow 6PbCl_2 + 4H_2O + Cl_2$	Reaction of red lead with HCl
20	$2KMnO_4 + 16HCl \rightarrow 2KCl + 2MnCl_2 + 8H_2O + 5Cl_2$	Oxidation of HCl by $KMnO_4$

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10 Diagrams

1. **(2019)** Electron dot structure of ammonium ion
2. **(2019)** Electron dot structure of nitrogen molecule
3. **(2020)** Electron dot diagram of calcium oxide (CaO)
4. **(2020)** Electron dot diagram of chlorine molecule (Cl_2)
5. **(2020)** Electron dot diagram of water (H_2O)
6. **(2022 Sem 2)** Structural diagram of 2-butene
7. **(2023)** Structural diagram of 2-chlorobutane
8. **(2023)** Structural diagram of methanal
9. **(2023)** Structural diagram of but-2-yne
10. **(2023)** Diagram of fountain experiment for HCl or NH_3

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10 Definitions

1. **Water of Crystallisation (IX Syllabus)**: The definite number of water molecules bound to some salts in their crystalline state.

2. **Efflorescence (IX Syllabus)**: The phenomenon where hydrated salts lose their water of crystallisation when exposed to air.

3. **Deliquescence (IX Syllabus)**: The process in which a substance absorbs moisture from the atmospheric air to become moist and ultimately dissolves in the absorbed water.

4. **Hygroscopy (IX Syllabus)**: The property of a substance to absorb moisture from the air without dissolving in it.

5. **Dehydrating Agent (IX Syllabus)**: A substance that removes chemically combined water from a compound.

6. **Drying Agent (IX Syllabus)**: A substance that removes moisture (physically combined water) from gases or liquids.

7. **Hard Water (IX Syllabus)**: Water that does not produce lather easily with soap due to the presence of dissolved calcium and magnesium salts.

8. **Soft Water (IX Syllabus)**: Water that produces lather easily with soap due to the absence of dissolved calcium and magnesium salts.

9. **Temporary Hardness (IX Syllabus)**: Hardness of water caused by the presence of dissolved bicarbonates of calcium and magnesium.

10. **Permanent Hardness (IX Syllabus)**: Hardness of water caused by the presence of dissolved chlorides and sulphates of calcium and magnesium.